

# Coding & Solution - OptiCrop (Smart Agriculture)

| Field                         | Details                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | <a href="https://github.com/amruthagutti1006-debug/Smart_Agriculture">https://github.com/amruthagutti1006-debug/Smart_Agriculture</a> |
| Programming Language(s)       | Python, HTML, CSS, JavaScript                                                                                                         |
| Framework(s) Used             | Flask, Scikit-learn, Pandas, NumPy                                                                                                    |
| Key Features Implemented      | Crop recommendation using soil and environmental parameters; ML prediction; User-friendly interface                                   |
| Pending / Incomplete Features | Weather API integration, fertilizer recommendation, multilingual support                                                              |
| Setup / Run Instructions      | Install requirements, run train_model.py (if needed), then execute app.py and open the local server                                   |

| S.No | Criteria                                 | Status |
|------|------------------------------------------|--------|
| 1    | Code is modular and organized            | Yes    |
| 2    | Meaningful variable/function names       | Yes    |
| 3    | Comments/documentation included          | Yes    |
| 4    | Error handling implemented               | Yes    |
| 5    | Application runs without critical errors | Yes    |
| 6    | Code committed to GitHub                 | Yes    |

**Additional Notes:** The OptiCrop project follows clean coding practices with a modular Flask architecture, reusable machine learning model, organized project structure, and GitHub version control.